

User Guide



WARESAFE

IoT-based Security System with Cybersecurity Simulation

Game Development | Cybersecurity | Internet of Things

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TABLE OF CONTENTS

WareSafe Product

01

Dashboard Guide

03

Cyber Attack Simulation Scenario

07

Game Simulation Guide

08

Features Guide

13

Welcome to WARESAFE !



WARESAFE

IoT-based Security System with Cybersecurity Simulation

Waresafe is an integrated solution for smart factory and warehouse security. This platform serves a dual purpose: it provides users with real-time monitoring of the physical workspace while offering an interactive cybersecurity training environment.

WARESAFE FEATURES



**Smart
Sensor
System**

Through our centralized dashboard, users can track live sensor data from secured areas, including RFID-based access control, motion detection in offices, and vibration monitoring for critical assets. By presenting all monitoring information in one place, the dashboard helps users observe system activity and respond to security events more efficiently.



**Cyber Attack
Simulation
Scenario**

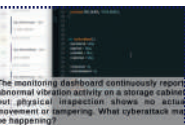
To help users better recognize and respond to modern digital threats, WareSafe includes a cybersecurity learning module that provides a safe environment for practicing cyber-attack scenarios. Users can explore interactive simulations and complete quiz-based learning activities without affecting the actual monitoring system. WareSafe is designed to keep both your physical and digital workplace safe!



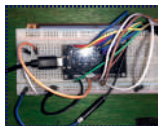
**Game
Simulation
for Education**



**Real-Time
Dashboard
Monitoring**



The monitoring dashboard continuously reports abnormal vibration activity on a storage cabinet. Not physical inspection shows no actual movement or tampering. What cyberattack may be happening?





WHICH ROLE ARE YOU?

SECURITY ADMIN

- Monitor all Office and Warehouse sensors
- Observe system activities and Alerts
- Review access and alert Logs
- Monitor and respond security incidents

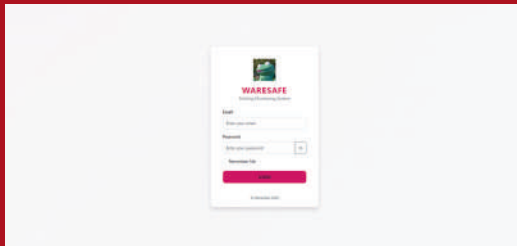
EMPLOYEE/USERS

- Access Warehouse and or Office Area
- Perform Daily Activities
- Interact with the system as part of normal operations

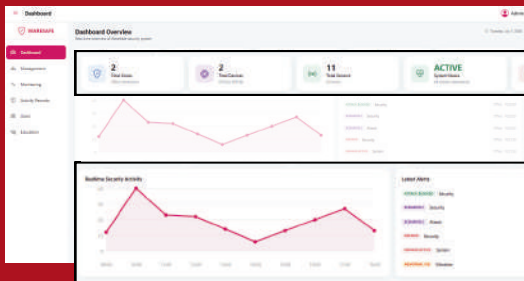


SECURITY ADMIN ROLE

- Login using Security Admin credentials



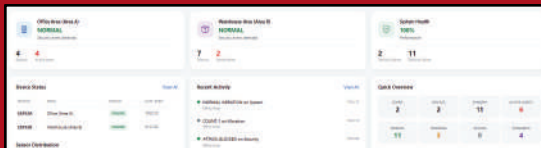
- After successfully login, you'll see Overview Page



Quick glance of everything

Latest Alert

Chart of security activity



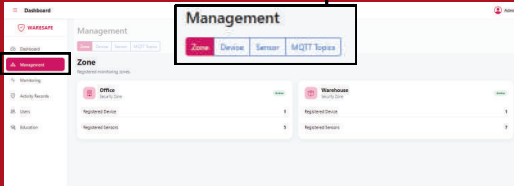
More detailed overview



SECURITY ADMIN ROLE

- Open Dashboard Management

Choose which management features to display



Device Management



Sensor Management

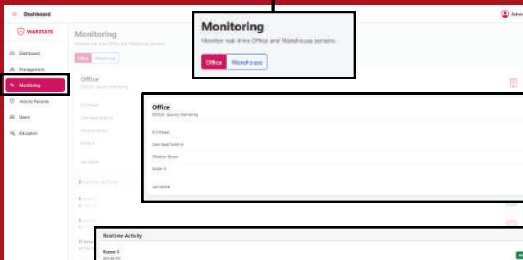


MQTT Topics Management



- Check Monitoring Panel

Select the Area Monitoring to view



Installed Sensor in the Area

Real Time Activity of the installed sensors



SECURITY ADMIN ROLE

- [Check Activity Records Panel](#)

Detailed records of normal activity are available here.

Cyber Attack Scenario

Defense Activity

Also for Defense Activity

Scroll more to view Cyber Attack Scenarios records

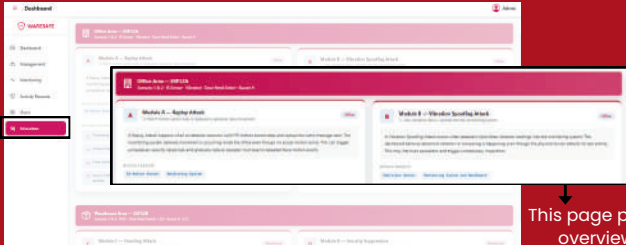
- [Go to Users Panel](#)

Manage staff names for each role and register warehouse access on this page



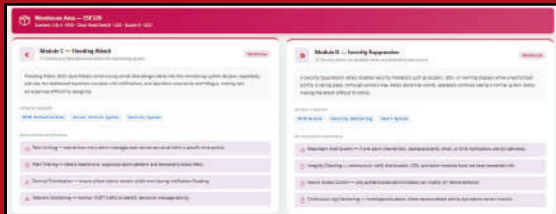
SECURITY ADMIN ROLE

- Read each module inside the Education Panel

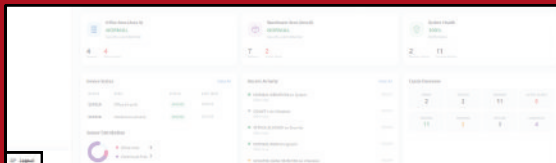


This page provides an overview of the cyberattack simulations to be tested

Warehouse Module



- Read each module inside the Education Panel



Logout to securely end the current session. After logging out, the system redirects to the login page



MALICIOUS INSIDER ATTACK SIMULATION

The following simulations assign different roles to represent various cyberattack scenarios. Each role performs specific actions to demonstrate the behavior and impact of the simulated attack on the WareSafe system.

Replay Attack Simulation

Role: Employee

Using previously captured sensor data, the Employee simulates legitimate system activity. This scenario highlights how replayed data can deceive the monitoring system.

Vibration Spoofing Attack Simulation

Role: Employee

To imitate abnormal warehouse conditions, the Employee introduces false vibration readings into the system. The simulation shows how manipulated sensor data can trigger false alarms.

Flooding Attack Simulation

Role: Employee

A large number of alarm messages are continuously generated by the Employee to overwhelm MQTT communication. This scenario illustrates the effect of excessive traffic on dashboard performance.

Security Suppression Attack Simulation

Role: Security Admin & Former Employee

Working together, the Security Admin suppresses security responses while the Former Employee attempts to access the warehouse with an unauthorized RFID card. This scenario highlights the risks of insider privilege misuse.



USER ROLE

- Open the installed game application



Click the "Play" Button to start the simulation

- Click Next until the end of the dialogue



There will be some dialogue to follow. Use the 'A' key to move left and the 'D' key to move right

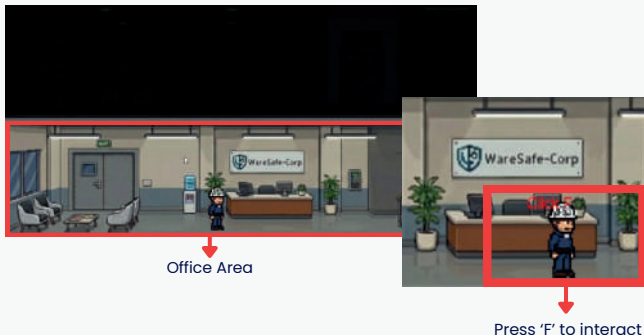


Move the character towards the office door to enter the Office Area



USER ROLE

- Once inside the Office Area, move the character to the computer to interact with it



- Select one of the available cyberattack scenarios



#1 Panel

This scenario does not include defense mechanisms



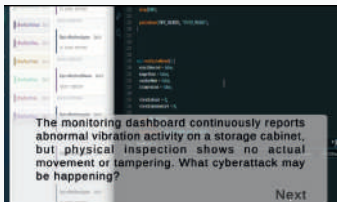
#2 Panel

This scenario includes added defense mechanisms



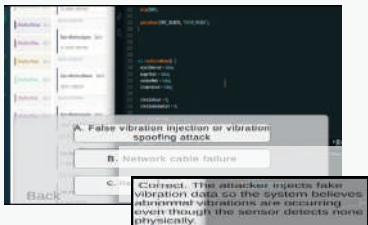
USER ROLE

- Watch the simulation video, then answer the quiz questions



#1 Quiz Panel

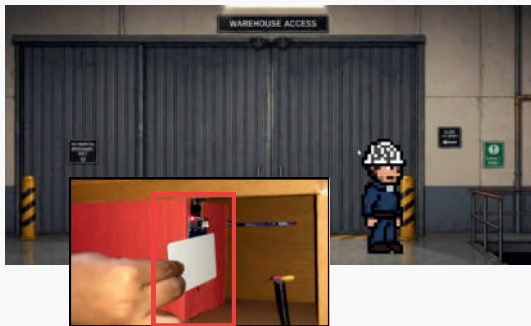
There are two question for each selected scenario



#2 Choices Answer Panel

There will be feedback for every answer

- After completing the quiz, choose whether to continue exploring or end the game. Selecting "Continue" proceeds to the Warehouse Area



Tap the RFID card on the RFID reader to gain access



USER ROLE

- After tapping the RFID card, enter the warehouse

Move the character near the computer and press 'F' to interact

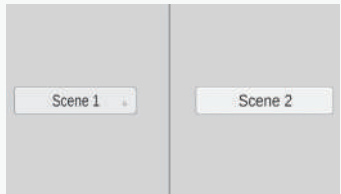


- Another list of cyberattack scenarios is displayed.
Select one of the available scenarios



#1 Panel

Similar with simulation in previous area, This scenario does not include defense mechanisms



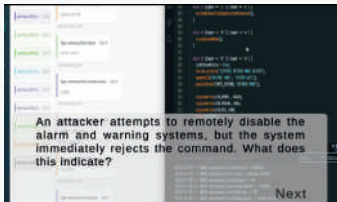
#2 Panel

This scenario includes added defense mechanisms



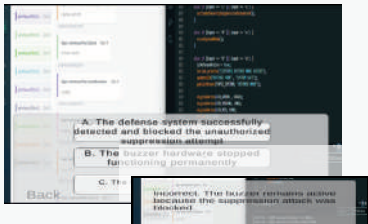
USER ROLE

- Watch the simulation video for the selected scenario.
After the video ends, answer the quiz questions



#1 Quiz Panel

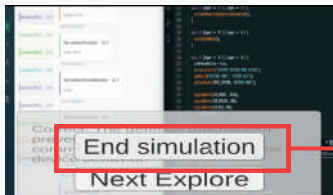
There are two question for each selected scenario



#2 Choices Answer Panel

There will be feedback for every answer

- After completing the quiz, choose whether to continue exploring or end the game



Select 'End Simulation' if you're done exploring!



TROUBLESHOOTING

Problem	Solution
Unable to log in	Ensure username and password are correct, contact the admin if needed
Dashboard not updating	Verify that the ESP32 is powered on and connected to Wi-Fi
Device appears offline	Check that the ESP32 and sensor modules are powered on, then make sure all wiring is securely connected
Simulation cannot start	Verify that the MQTT broker is running and the Unity application is connected to the broker
Quiz does not load	Restart the application and try again the simulation
MQTT disconnected	Make sure all system are connected to the same broker and network, then reconnect the MQTTX

LIMITATION

- Requires a stable Wi-Fi and MQTT connection
- Cyber attack simulations are intended for educational and demonstration purposes only
- Designed for deployment within a local network environment
- Additional customization may be required for larger-scale deployments

THANK YOU



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